數據、預測與 R^2

Data, predictions and R^2

班別 / Class: _____ 組別 / Group: _____ 日期 / Date: _____

程式在 Colab：儲存副本，依次執行。將表格結果與觀察記在這裏。

Save a Colab copy and run the cells in order. Record results and observations here.

1 數據探索：外送距離與收費

Data exploration: delivery distance and fee

模擬紀錄；每點是一筆外送紀錄。執行第 1 節，看看散點圖。

Simulated records; each dot is one delivery. Run Section 1 and inspect the scatter plot.

距離 / Distance (km)	1	2	3	4	5	6	7	8
收費 / Fee (HK$)	24	29	27	38	35	44	42	51

1. 距離增加時，收費大致怎樣變化？

As distance increases, how does the fee generally change?

2. 距離較遠，收費是否每次都較高？用兩筆紀錄說明。

Is the fee always higher for a longer distance? Use two records to explain.

2 線性回歸：溫習時間與成績

Linear regression: study time and exam score

執行第 2 節。將模型表的數值抄在下表；留意圖中的紫色預測點。

Run Section 2. Copy the model-table values below; find the purple prediction point.

項目 Metric	Colab 數值 Colab value	意思 Meaning
n	_____	觀察數目 Number of observations
m	_____	每多一小時，估計分數的改變 Change in estimated marks per extra hour
c	_____	0 小時的估計分數 Estimated marks at 0 hours
R^2	_____	直線解釋了多少數據差異 Share of variation explained by the line
4.5 小時預測 Prediction at 4.5 h	_____ 分 / marks	模型給出的估計，並非保證 An estimate, not a guarantee

m、c 描述直線；不是準確度指標。所有成績爲模擬。R^2 說明在背面。

m/c describe the line, not accuracy. Scores are simulated. See reverse for R^2.

Colab: gordany.github.io/f2-linear-regression/workshop-2/

R^2（R square）：直線解釋了多少數據差異？

R^2: how much variation in these data does the line explain?

這裏的「差異」指各個 y 值有高有低，例如學生的成績不同。

Variation means that y-values differ, e.g. students have different scores.

本課的 R^2：在用來建立模型的數據上計算，值介乎 0 和 1。

Here, R^2 is measured on the data used to fit the line and ranges from 0 to 1.

接近 1：解釋較多差異；接近 0：直線解釋的差異較少。

Near 1: more variation explained. Near 0: little variation explained.

例如 0.80：解釋約 80% 的差異，不是 80% 的預測答對。

For example, 0.80 means about 80% of variation, not 80% correct predictions.

較高 R^2 表示這組數據的擬合較好；新預測是否可靠，仍需用新數據檢查。

Higher R^2 means a better fit here. Reliable new predictions still need testing on new data.

觀察太少，信心有限；關聯亦不能證明因果。

Few observations limit confidence; association does not prove cause.

3 比較 A、B、C：哪條線更能解釋自己的數據？

Compare A, B and C: which line explains its data better?

在第 3 節依次執行 A、B、C；各有一個程式格、21 點及相同座標範圍。

Run A, B and C in Section 3. Each has one code cell, 21 dots and the same axes.

先看圖，再記下 R^2。第 1 名代表解釋自己數據差異的比例最高。

Inspect each graph and record R^2. Rank 1 means the most variation explained.

例子 Example	R^2（抄至 3 位小數） R^2 (copy 3 decimal places)	排名：1 / 2 / 3 Rank: 1 / 2 / 3
A	_____	_____
B	_____	_____
C	_____	_____

1. 第一名的圖中，你看到甚麼支持這個排名？

What do you see in the top-ranked graph that supports your ranking?

2. 高 R^2 是否保證能準確預測新數據？還需要做甚麼檢查？

Does high R^2 guarantee accurate new predictions? What check is still needed?

三組都是模擬數據。隨機數據也可能碰巧有小幅趨勢。 / Random data can show a chance trend.